📊 Email Generation Assistant Report

This report presents the design, implementation, evaluation, and comparative analysis of an AI-powered Email Generation Assistant. The system generates professional, structured emails based on user inputs — **Intent, Key Facts, and Tone —** using modern LLMs.

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1. Prompt Template

The system uses an **advanced prompt engineering strategy** designed to improve output consistency and quality.

Prompting Techniques Used

- **Role-based prompting**
- **Few-shot example guidance**

Prompt Design

The model is explicitly assigned the role:

> "World-class Executive Assistant and Professional Email Writer"

This ensures that outputs are:

- Professionally structured
- Context-aware
- Tone-consistent

The model is guided to:

- Generate complete emails (Subject, Greeting, Body, Closing)
- Integrate all key facts naturally (no bullet dumping)
- Maintain consistent tone throughout the email

Few-shot Example

A reference example is embedded in the prompt showing:

Component	Description
Input	Intent + Facts + Tone
Output	Fully structured professional email

This significantly improves:

- Output structure
- Fact integration
- Tone alignment

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2. Custom Evaluation Metrics

A custom evaluation framework was implemented using an **LLM-as-a-Judge** approach.

Metric	Definition	Evaluation Logic	Range
Relevance (Fact Recall)	Measures inclusion of key facts	LLM verifies presence of facts	0–1
Professional Quality	Grammar, tone, structure	LLM rating (1–5 → normalized)	0–1
Personalization	Specific vs generic output	LLM rating (1–5 → normalized)	0–1

3. Model Comparison and Analysis

Models Evaluated

- **Model A:** `llama-3.1-8b-instant` (fast baseline)
- **Model B:** `openai/gpt-oss-120b` (high-quality model)

📊 Performance Summary

Metric	Model A	Model B
Relevance	0.84	0.96
Quality	0.72	0.93
Personalization	0.62	0.91
Overall Score	**0.73**	**0.93**

Observations from Outputs

🔴 Model A (llama-3.1-8b-instant)

- Occasionally produced **incomplete or loosely structured emails**
- Sometimes **missed subject lines**
- Responses were **more generic and repetitive**
- Less consistent formatting across scenarios

🟢 Model B (openai/gpt-oss-120b)

- Consistently generated **well-structured emails**
- Always included:
  - Subject line
  - Clear formatting
  - Professional tone
- Demonstrated:
  - Stronger **context understanding**
  - Better **fact integration**
  - Higher **personalization quality**

Example improvements observed:

- Included quantified achievements
- Used more refined professional language
- Maintained tone consistently across the email

Key Failure Mode (Model A)

The primary limitations observed:

- Weak structural consistency (missing subject/formatting)
- Generic and less engaging content
- Inconsistent tone control

Final Recommendation

I recommend **openai/gpt-oss-120b** for production use because:

- Superior email structure
- Better personalization and contextual depth
- Stronger tone consistency
- More polished and professional output

While Model A offers faster responses, the quality difference is significant for real-world applications.

4. Evaluation Data

Evaluation was conducted on **10 scenarios**, each containing:

- Intent
- Key Facts
- Tone

Raw results are available in:

- `evaluation/results_model_a.csv`
- `evaluation/results_model_b.csv`

Each file contains:

- Scenario ID
- Metric scores (Relevance, Quality, Personalization)
- Average score per scenario
- Overall model average

5. Conclusion

This project demonstrates:

- Effective use of prompt engineering techniques
- A task-specific evaluation framework for email generation
- A clear performance gap between smaller and larger models

The results show that **higher-capacity models significantly improve output usability**, especially in structured communication tasks like professional email generation.